

DESIGN AND IMPLEMENTATION OF A SECURITY OPERATIONS CENTER (SOC) LAB USING **WAZUH SIEM**

*A Hands-on Security Operations Center (SOC) Lab
for Threat Detection, Monitoring,
and Incident Investigation*



TOOLS & TECHNOLOGIES USED

-  Wazuh SIEM (v4.14.5)
-  Docker Desktop
-  Windows 10 Virtual Machine
-  Sysmon
-  VirtualBox
-  PowerShell
-  Nmap
-  File Integrity Monitoring (FIM)
-  Threat Hunting
-  MITRE ATT&CK Framework
-  OpenSearch Dashboard
-  XML Rule Configuration



PREPARED BY

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Acknowledgement

I would like to express my sincere gratitude to everyone who contributed, directly or indirectly, to the successful completion of this project titled "**Design and Implementation of a Security Operations Center (SOC) Lab Using Wazuh SIEM.**"

This project provided me with valuable hands-on experience in Security Information and Event Management (SIEM), endpoint monitoring, threat detection, and digital forensics. Throughout the implementation, I explored the practical deployment of a Wazuh-based SOC environment using Docker, integrated Sysmon with a Windows endpoint, configured File Integrity Monitoring (FIM), created custom detection rules, and investigated security events using the Wazuh Dashboard.

I also acknowledge the availability of open-source cybersecurity tools and official documentation that supported the successful implementation of this project. These resources helped me gain practical knowledge of SOC operations, security monitoring, log analysis, threat hunting, and the MITRE ATT&CK framework.

Finally, I would like to thank everyone who has supported my learning journey and motivated me to continuously improve my cybersecurity skills. This project represents an important milestone in strengthening my practical understanding of Security Operations Center (SOC) workflows and preparing me for future opportunities in Cybersecurity and Digital Forensics.

Muskan Paraswani

Abstract

Security Operations Centers (SOCs) play a critical role in protecting organizations from modern cyber threats by continuously monitoring, detecting, and responding to security incidents. This project presents the design and implementation of a practical SOC laboratory using the **Wazuh Security Information and Event Management (SIEM) platform** to simulate real-world security monitoring and threat detection.

The SOC lab was deployed using **Docker** on a macOS host, while a **Windows 10 Virtual Machine** was configured as the monitored endpoint. **Sysmon** was installed to generate detailed Windows event logs, which were forwarded to the Wazuh Manager for analysis. The project demonstrates the integration of Wazuh agents, centralized log collection, event correlation, File Integrity Monitoring (FIM), and threat hunting using the Wazuh Dashboard.

Several security scenarios were performed to evaluate the effectiveness of the SOC environment, including process creation monitoring, PowerShell activity detection, Windows service creation monitoring, file integrity monitoring, and custom rule development. A custom Wazuh detection rule was successfully created to identify the execution of the **Net User** command, demonstrating the ability to extend Wazuh's default detection capabilities. Security events were analyzed using the MITRE ATT&CK framework, enabling better understanding of attacker tactics and techniques.

The implementation highlights how open-source cybersecurity tools can be used to build a functional Security Operations Center for monitoring endpoint activity, detecting suspicious behavior, and performing security investigations. This project strengthens practical skills in SIEM deployment, log analysis, threat hunting, incident monitoring, and defensive cybersecurity operations, making it an effective learning platform for aspiring SOC analysts and digital forensics professionals.

1. Introduction

In today's rapidly evolving digital landscape, cyber threats have become increasingly sophisticated and frequent, making continuous security monitoring an essential requirement for organizations. A **Security Operations Center (SOC)** is a centralized facility responsible for monitoring, detecting, analyzing, and responding to cybersecurity incidents in real time. SOC analysts use Security Information and Event Management (SIEM) platforms to collect logs from multiple systems, correlate events, identify suspicious activities, and support incident response.

Wazuh is a powerful open-source SIEM and Extended Detection and Response (XDR) platform that provides log analysis, file integrity monitoring, vulnerability detection, threat hunting, security configuration assessment, and incident detection. Due to its flexibility and open-source nature, Wazuh is widely used in cybersecurity laboratories, educational environments, and enterprise security operations.

The primary objective of this project is to design and implement a practical SOC laboratory using Wazuh to simulate real-world security monitoring. The environment was deployed using **Docker** on a macOS host, while a **Windows 10 virtual machine** was configured as the monitored endpoint. **Sysmon (System Monitor)** was installed on the Windows endpoint to generate detailed system activity logs, enabling enhanced visibility into process creation, PowerShell execution, service creation, and other security-relevant events.

During the implementation, multiple security monitoring scenarios were performed, including process monitoring, PowerShell activity detection, Windows service monitoring, File Integrity Monitoring (FIM), and the development of a custom Wazuh detection rule. The generated security events were analyzed using the Wazuh Dashboard and mapped to the **MITRE ATT&CK** framework to better understand attacker tactics and techniques.

This project demonstrates the practical deployment of an open-source Security Operations Center and provides hands-on experience with SIEM implementation, threat detection, endpoint monitoring, log analysis, and defensive cybersecurity operations. The knowledge gained through this project strengthens the practical skills required for roles such as SOC Analyst, Cybersecurity Analyst, and Digital Forensics Investigator.

2. Project Objectives

The primary objective of this project is to design and implement a fully functional **Security Operations Center (SOC) laboratory** using the **Wazuh SIEM** platform to gain practical experience in cybersecurity monitoring, threat detection, and incident analysis.

The specific objectives of this project are:

- To deploy the Wazuh SIEM platform using Docker containers.
- To configure a Windows 10 virtual machine as an endpoint and install the Wazuh agent.
- To integrate Sysmon with Wazuh for enhanced Windows event logging and security monitoring.
- To collect, centralize, and analyze endpoint logs through the Wazuh Dashboard.
- To monitor system activities such as process creation, PowerShell execution, Windows service creation, and authentication events.
- To implement File Integrity Monitoring (FIM) for detecting unauthorized file changes.
- To create and test custom Wazuh detection rules for identifying suspicious activities.
- To investigate generated security alerts and understand the corresponding attack techniques.
- To map detected events with the MITRE ATT&CK Framework for better threat classification.
- To develop practical skills in SIEM deployment, threat hunting, incident monitoring, and defensive cybersecurity operations.
- To build a professional cybersecurity portfolio project demonstrating hands-on SOC analyst capabilities.

3. Lab Architecture / Environment Setup

The SOC Lab environment is designed to simulate a real-world Security Operations Center workflow where security events are generated, collected, monitored, and analyzed.

The lab consists of:

Host Machine

- Personal computer system used for running the lab environment

Security Analysis Machine

- Kali Linux
- Used for security testing, investigation, and analysis activities

Monitoring Platform

- Wazuh SIEM
- Used for collecting logs, generating alerts, and monitoring security events

Network Monitoring

- Wireshark
- Used for capturing and analyzing network traffic

The workflow of the SOC Lab:

1. Generate security events
2. Collect logs from systems
3. Monitor events through SIEM
4. Analyze alerts
5. Perform investigation
6. Document incident findings

4. Installation and Configuration

Installation and Configuration

The SOC Lab environment was prepared by installing and configuring different security monitoring components required for log collection, detection, and analysis.

1. Docker Installation

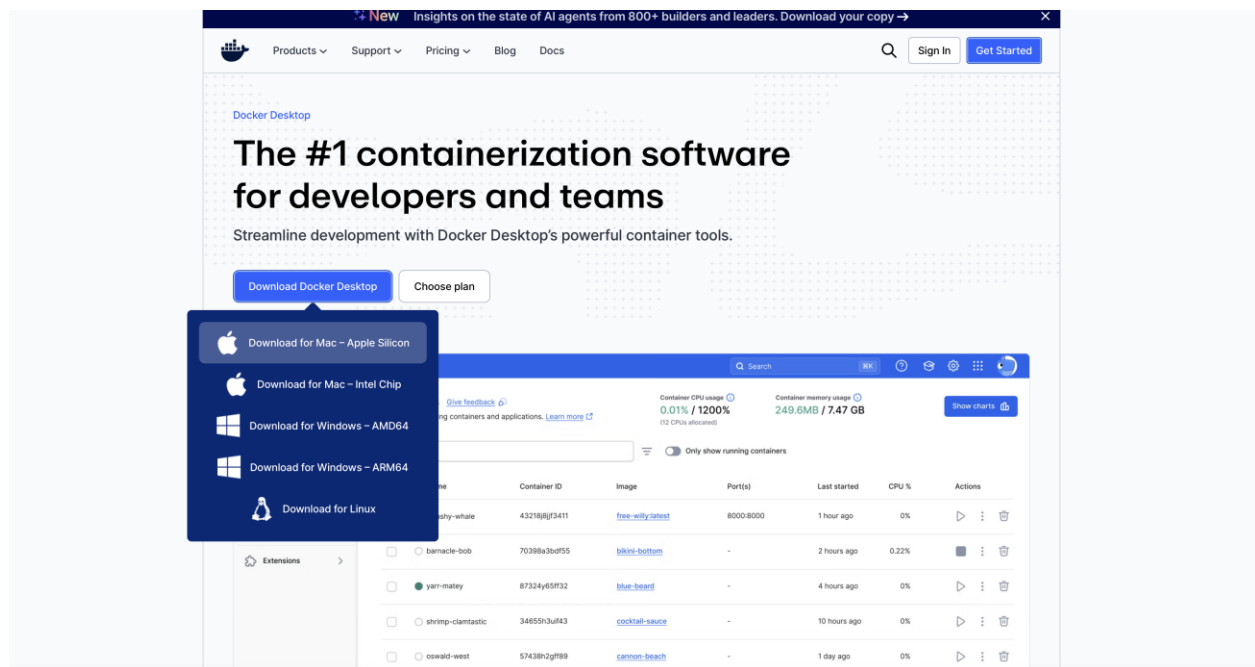
Docker was installed to deploy and manage the SOC monitoring environment efficiently.

Docker provides a containerized environment that allows security tools to run with proper isolation and easier management.

Docker Setup Process:

- Installed Docker on the host system
- Verified Docker installation
- Configured Docker services required for SOC tools
- Used Docker containers for deploying Wazuh components

Screenshots:



```
muskanrajaraswani@Mac Downloads % docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

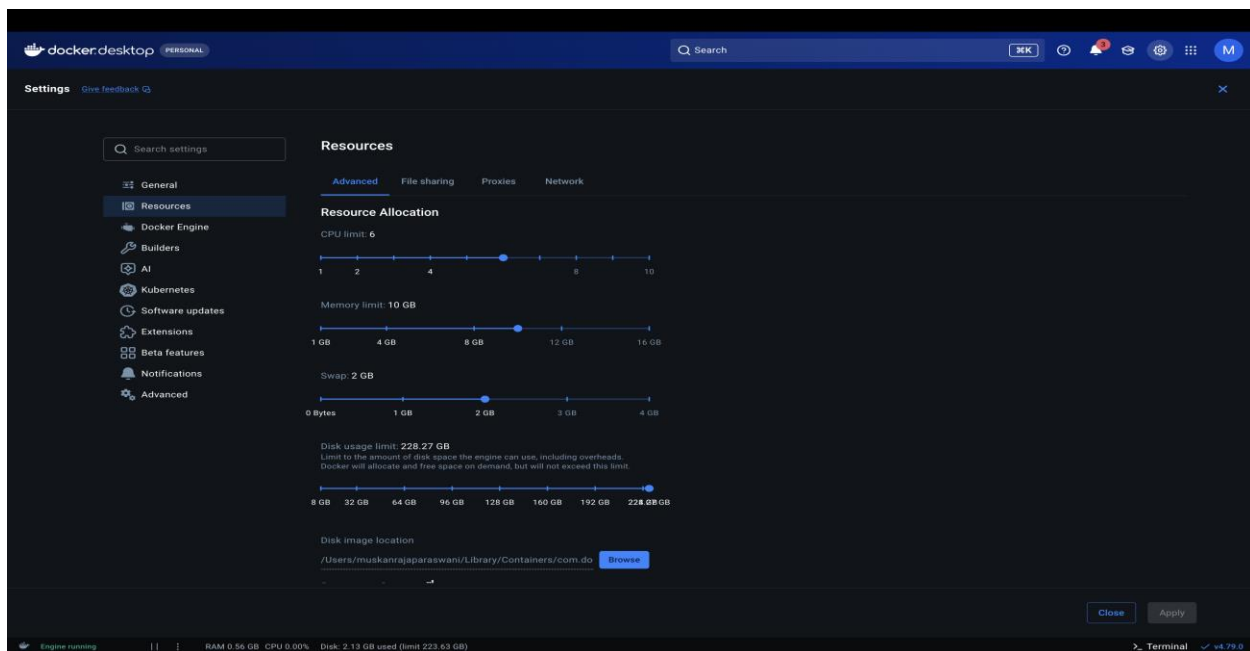
To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (arm64v8)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

muskanrajaraswani@Mac Downloads %
```



2. Wazuh SIEM Deployment

Wazuh was deployed using Docker containers to create a Security Information and Event Management (SIEM) environment.

Wazuh was used for:

- Collecting endpoint logs
- Monitoring security events
- Generating alerts
- Detecting suspicious activities
- Performing security analysis

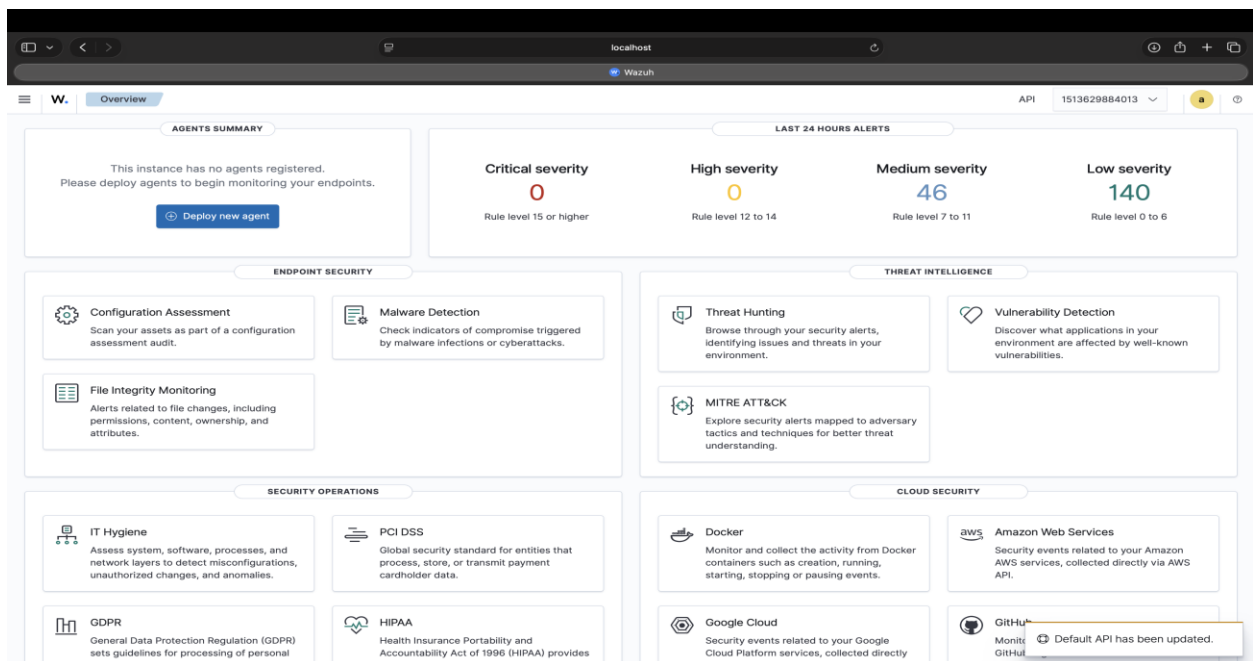
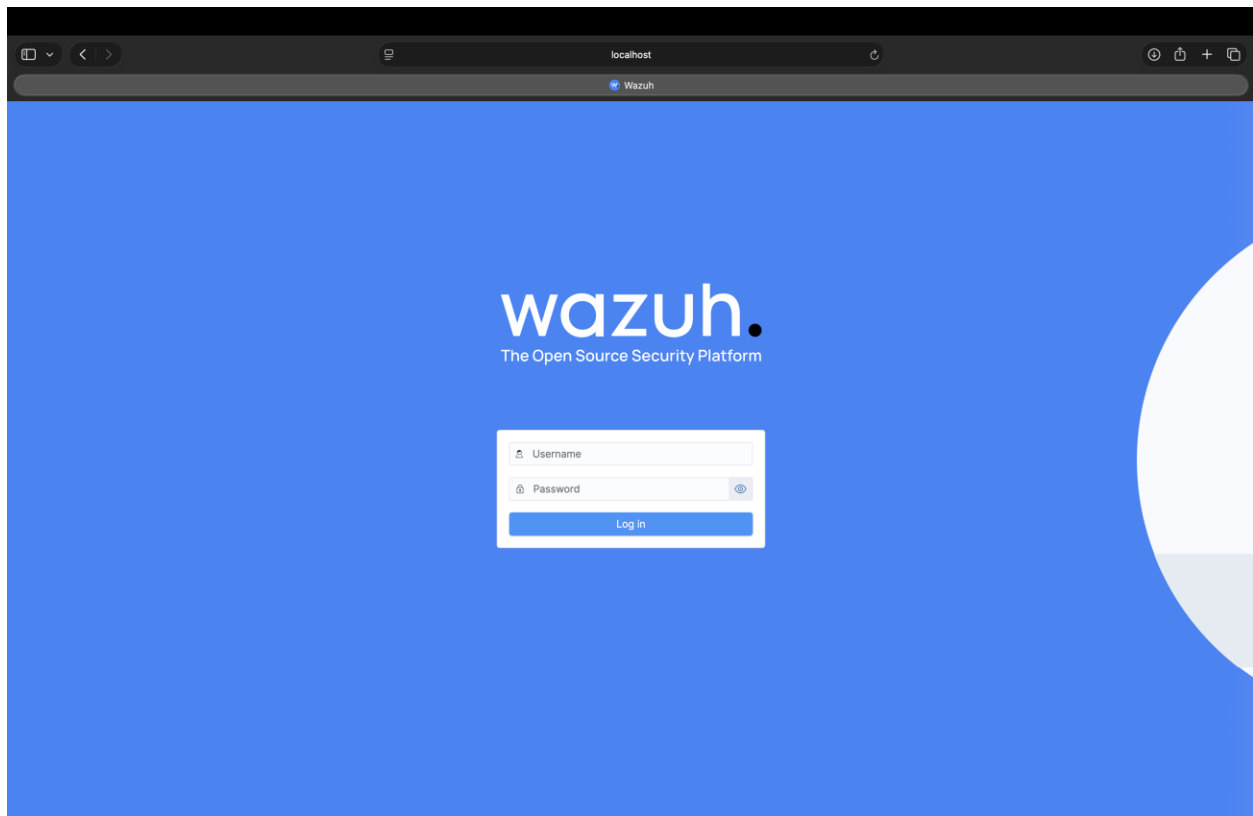
Wazuh Components:

- Wazuh Manager – Processes and analyzes security events
- Wazuh Dashboard – Provides visualization and alert monitoring
- Wazuh Indexer – Stores and searches collected data

Screenshots:

```
muskanrajaparawani@Mac single-node % docker pull wazuh/wazuh-manager:4.14.5
4.14.5: Pulling from wazuh/wazuh-manager
1020671f3c5f: Pull complete
4c954a8005ab: Pull complete
92ab589d560b: Pull complete
be6f5a4e2de1: Pull complete
4f4fb70ef54: Pull complete
04550bb8f16a: Pull complete
11d1632cb4cf: Pull complete
f01e84c48cba: Pull complete
6c159a10e412: Pull complete
44209164e8d5: Pull complete
cfbd825507ff: Pull complete
8c58456ea52e: Pull complete
be978697b103: Pull complete
7f70592f628f: Pull complete
e834ab867f34: Pull complete
1126dbb15fd6: Pull complete
f5422524217a: Download complete
Digest: sha256:824ff13d446682cd3bd5960479ce0d1d6a9fed8fdb0ce31a4f0f3b9a97a7734
Status: Downloaded newer image for wazuh/wazuh-manager:4.14.5
docker.io/wazuh/wazuh-manager:4.14.5
muskanrajaparawani@Mac single-node %
```

```
muskanrajaparawani@Mac single-node % docker compose up -d
[+] up 3/3
✓ Container single-node-wazuh-manager-1 Started 0.2s
✓ Container single-node-wazuh-indexer-1 Started 0.2s
✓ Container single-node-wazuh-dashboard-1 Started 0.2s
muskanrajaparawani@Mac single-node % docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
6db28557eb43   wazuh/wazuh-dashboard:4.14.5       "/entrypoint.sh"        7 seconds ago Up 6 seconds  0.0.0.0:443->5681/tcp
single-node-wazuh-dashboard-1
64c1d747030   wazuh/wazuh-indexer:4.14.5         "/entrypoint.sh open-"  7 seconds ago Up 6 seconds  0.0.0.0:9200->9200/tcp
single-node-wazuh-indexer-1
b4a5e081acdf   wazuh/wazuh-manager:4.14.5         "/init"                 7 seconds ago Up 6 seconds  0.0.0.0:1514->1514-1515/tcp, [::]:1514->1514-1515/tcp, 0.0.0.0:514->514/udp, [::]:514->514/udp, 0.0.0.0:55000->55000/tcp, [::]:55
single-node-wazuh-manager-1
muskanrajaparawani@Mac single-node %
```



3. Windows VM and Sysmon Setup

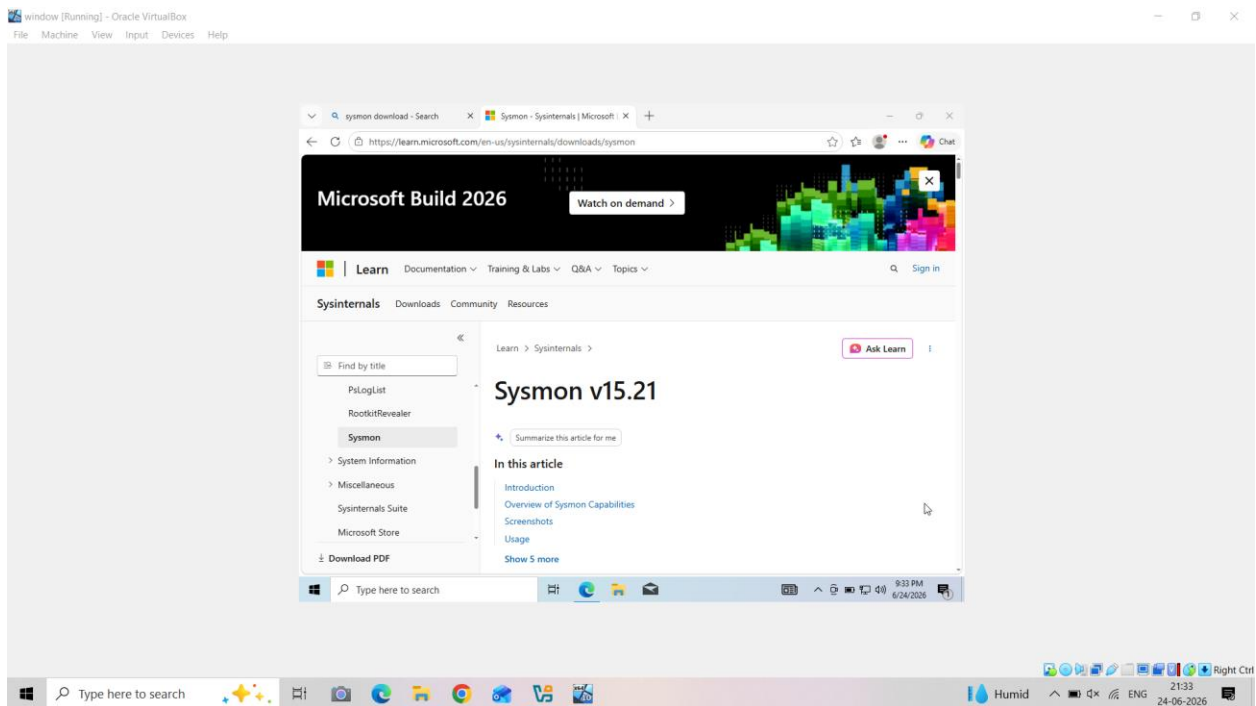
A Windows Virtual Machine was configured as an endpoint system for generating and collecting security events.

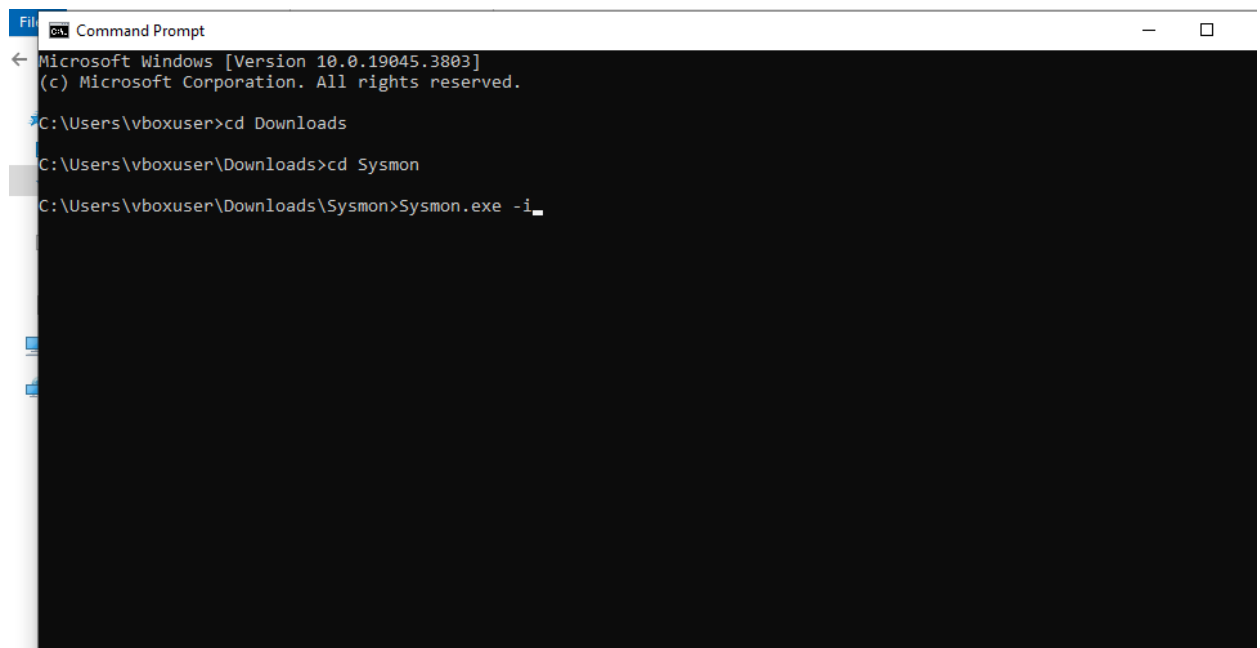
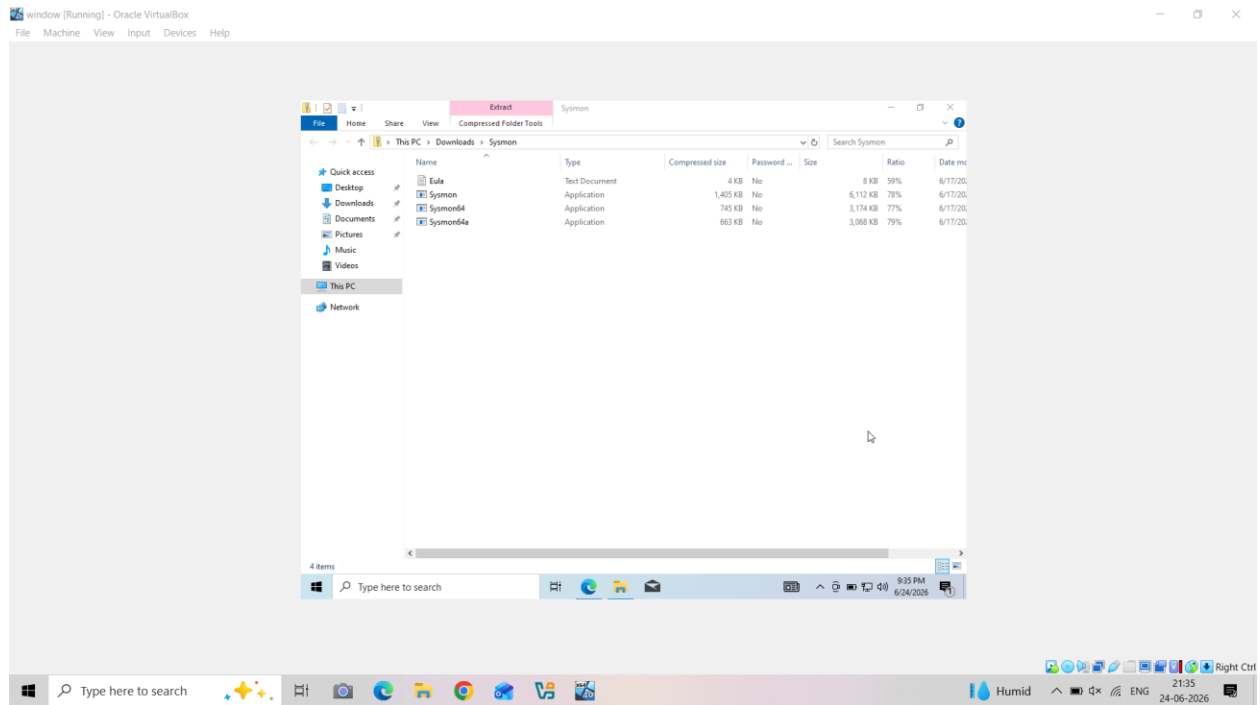
Sysmon (System Monitor) was installed on Windows VM to capture detailed system activities.

Sysmon monitors:

- Process creation
- Network connections
- File activities
- Registry modifications
- System events

Screenshots:





4. Log Collection Flow

The final SOC Lab workflow:

Windows VM



Sysmon Generates Events



Wazuh Agent Collects Logs



Wazuh Manager Analyzes Events



Dashboard Displays Alerts



SOC Analyst Investigates Incident

5. Attack Simulation & Detection Process

To test the SOC monitoring environment, different security events were generated and analyzed.

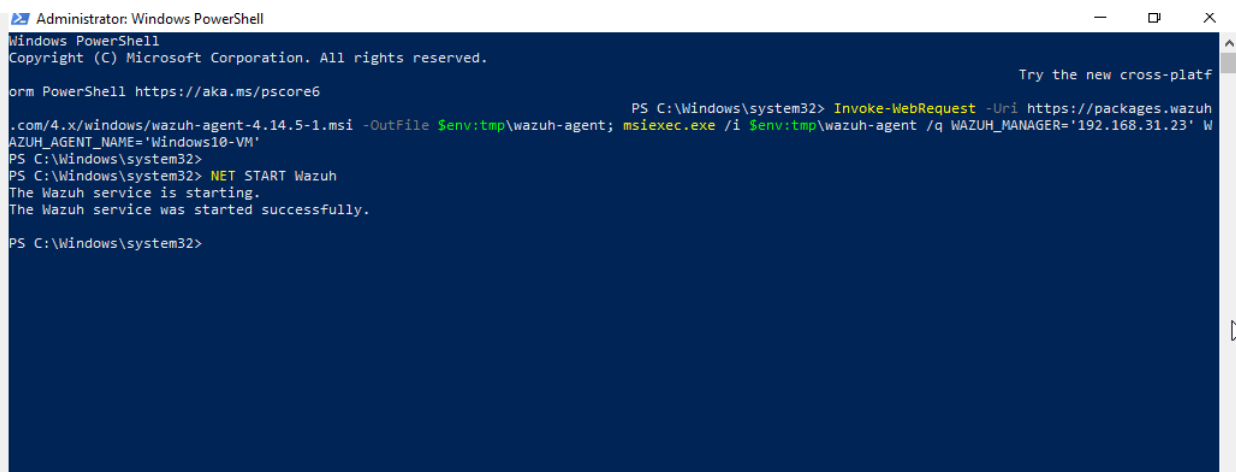
The purpose of attack simulation was to understand how a SOC analyst detects suspicious activities and investigates security alerts.

Simulation Activities

- Multiple failed login attempts were generated to simulate a brute-force attack.
- Network traffic was monitored to identify unusual communication patterns.
- System logs were analyzed to detect abnormal activities.
- Security alerts generated by the monitoring platform were reviewed.

Detection Workflow

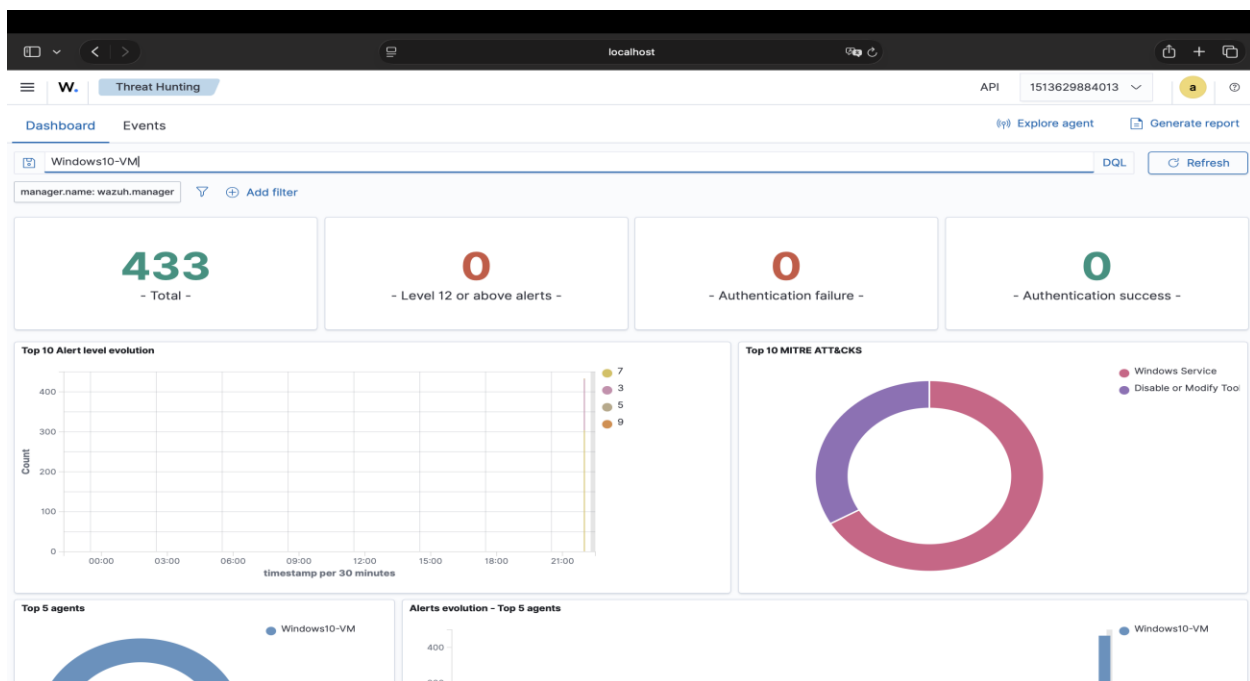
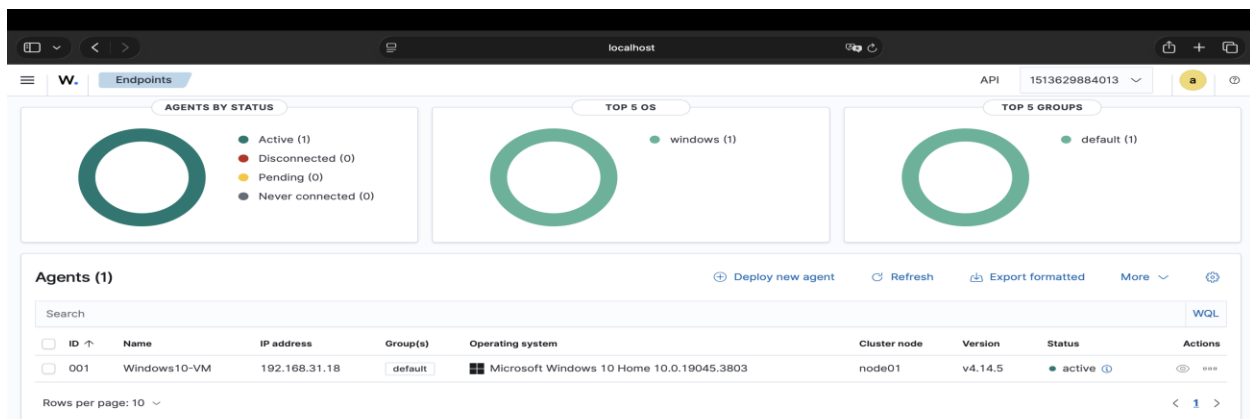
1. Security event occurs on the system.
2. Logs are generated by the operating system.
3. Logs are collected by the SIEM platform.
4. SIEM analyzes the activity and generates alerts.
5. Analyst investigates the alert and identifies the threat.
6. Incident details are documented for response actions.

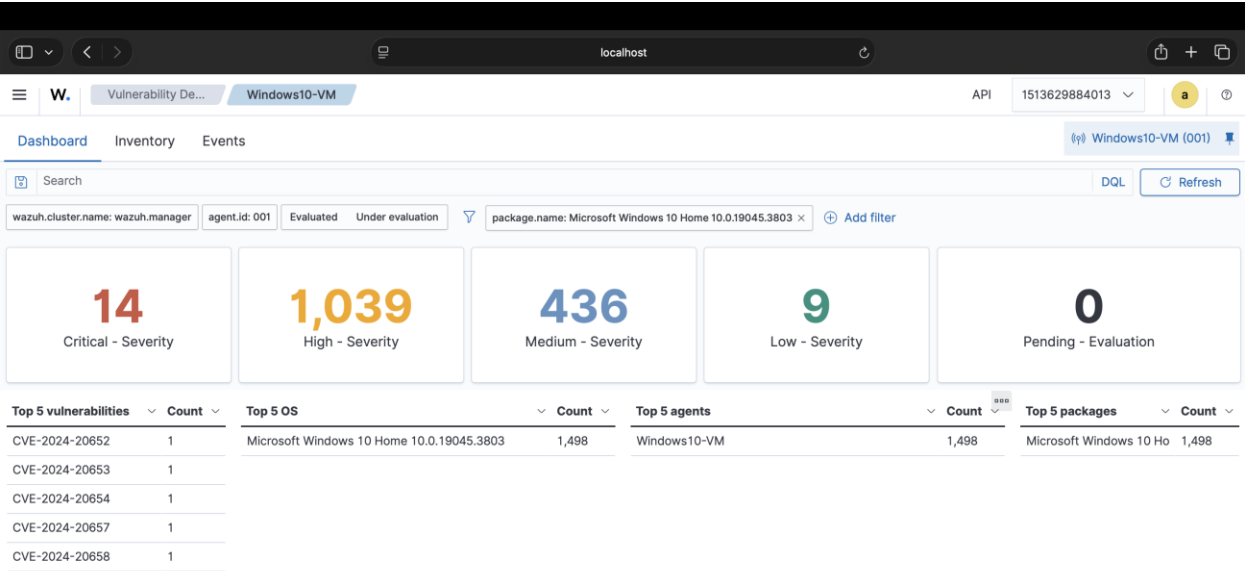
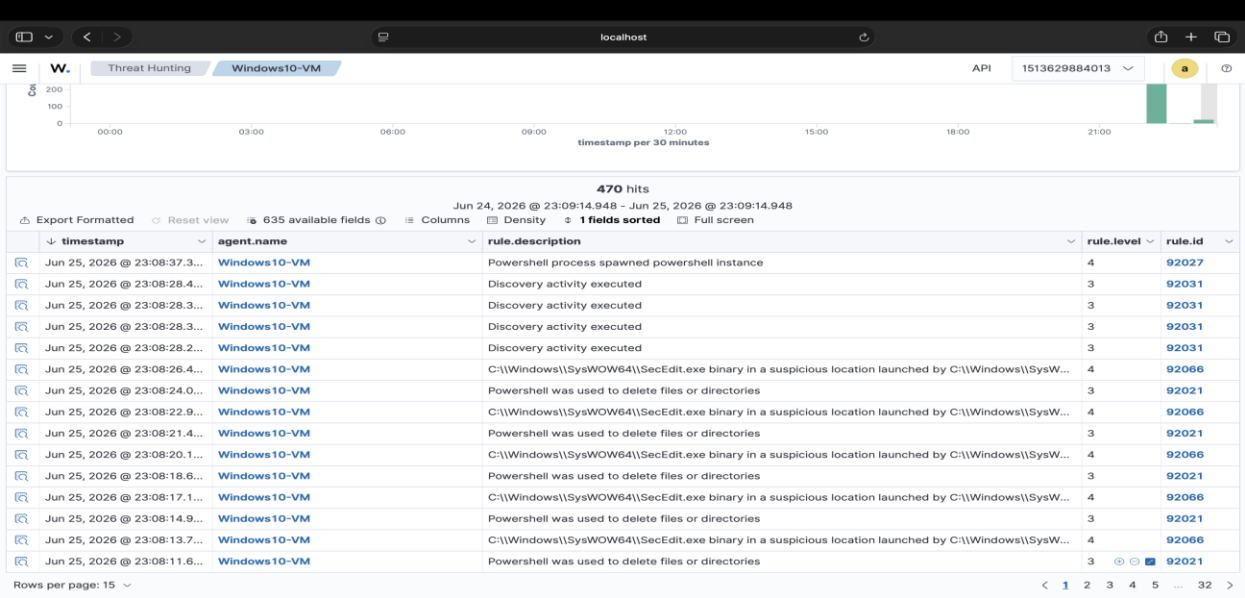


```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.5-1.msi -OutFile $env:tmp\wazuh-agent; msexec.exe /i $env:tmp\wazuh-agent /q WAZUH_MANAGER='192.168.31.23' W
AZUH_AGENT_NAME='Windows10-VM'
PS C:\Windows\system32>
PS C:\Windows\system32> NET START Wazuh
The Wazuh service is starting.
The Wazuh service was started successfully.
PS C:\Windows\system32>
```



6.Alert Monitoring and Analysis

The SOC monitoring process involves reviewing security alerts generated by the SIEM platform.

During analysis, the following information was examined:

- Alert type and severity level
- Source IP address
- Target system details
- Time and frequency of events
- Related log entries
- Possible security impact

The investigation process helps determine whether an alert is:

- True Positive – Actual security threat detected
- False Positive – Normal activity incorrectly identified as a threat

Based on the analysis, appropriate incident response actions are performed.

7. Custom Detection Rules Configuration

To improve threat detection capability, custom detection rules were created in the Wazuh SIEM environment.

Custom rules help the SOC analyst identify specific security events based on collected logs and trigger alerts when suspicious activities occur.

Rule Creation Process

The custom rules were created by modifying the Wazuh rules configuration file.

The rules were designed to detect:

- Suspicious login activities
- Multiple failed authentication attempts
- Unusual system behavior
- Security-related events generated by Sysmon

Rule Implementation

Steps followed:

1. Accessed Wazuh manager configuration.
2. Created custom rules inside the Wazuh rules directory.
3. Defined rule conditions based on log events.
4. Assigned rule IDs and severity levels.
5. Restarted Wazuh manager to apply changes.
6. Generated test events to verify detection.

Custom Rule Example

Rule Purpose:

Detect suspicious authentication activity.

Detection Logic:

- Monitor failed login attempts
- Identify repeated authentication failures
- Generate security alert when threshold is reached

Screenshots:

```
<!-- Local rules -->
<!-- Modify it at your will. -->
<!-- Copyright (C) 2015, Wazuh Inc. -->
<!-- Example -->
<group name="local.syslog.sshd">
  <!--
  Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1 port 1000 ssh2
  -->
  <rule id="100001" level="5">
    <if_sid>5710</if_sid>
    <timestamp>1.1.1.1</timestamp>
    <description>SSH Authentication failed from IP 1.1.1.1</description>
    <group>authentication_failed<ACL>000-100-100-100-100-100-100-100</group>
  </rule>

  <!-- Custom Rule -->
  <rule id="100100" level="12">
    <if_sid>5710</if_sid>
    <match>net user</match>
    <description>Custom Rule: User Enumeration detected using 'net user'</description>
    <group>custom.user_enumeration<ACL>100-100-100</group>
  </rule>
</group>
```

W.

Threat Hunting

Windows10-VM

API

1513629884013

a

ⓘ

Dashboard

Events

🔍 Windows10-VM (001)

🔍 Search

DQL

📅 Last 24 hours

Show dates

🔄 Refresh

manager.name: wazuh.manager

agent.id: 001

🔍 Add filter

timestamp per 30 minutes	Count
22:00	447

447 hits

Jun 24, 2026 @ 22:40:02.955 - Jun 25, 2026 @ 22:40:02.955

📄 Export Formatted

🔄 Reset view

📊 635 available fields

📄 Columns

📊 Density

📄 1 fields sorted

🖥 Full screen

↓ timestamp	agent.name	rule.description	rule.level	rule.id
🔍 Jun 25, 2026 @ 22:39:11.5...	Windows10-VM	Logon Failure - Unknown user or bad password	5	60122
🔍 Jun 25, 2026 @ 22:39:04.5...	Windows10-VM	Logon Failure - Unknown user or bad password	5	60122

	Jun 26, 2026 @ 21:36:27.3...	Windows10-VM	Custom Rule: User Enumeration detected using 'net user'	12	100100
	Jun 26, 2026 @ 21:36:27.2...	Windows10-VM	Custom Rule: User Enumeration detected using 'net user'	12	100100
	Jun 26, 2026 @ 21:36:27.2...	Windows10-VM	Custom Rule: User Enumeration detected using 'net user'	12	100100
	Jun 26, 2026 @ 21:36:27.1...	Windows10-VM	Custom Rule: User Enumeration detected using 'net user'	12	100100

localhost

SOC Lab ProjectWazuh

W. Threat Hunting

API1513629884013a

DashboardEvents

Explore agent

Search

DQLLast 15 minutesShow datesUpdate

manager.name: wazuh.manager

Add filter

537 hits

Jun 25, 2026 @ 12:49:04.959 - Jun 26, 2026 @ 12:49:04.959

Export FormattedReset view717 available fieldsColumnsDensity1 fields sortedFull screen

timestamp	agent.name	rule.description	rule.level	rule.id
Jun 26, 2026 @ 12:47:31.3...	Windows10-VM	CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Ensure 'Enforce password history' is set to '24 ...	3	19009
Jun 26, 2026 @ 12:47:31.3...	Windows10-VM	CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Ensure 'Maximum password age' is set to '365 ...	3	19009
Jun 26, 2026 @ 12:47:04.2...	Windows10-VM	SCA summary: CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Score less than 30% (27)	9	19005
Jun 26, 2026 @ 12:46:47.3...	Windows10-VM	CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Ensure 'Maximum password age' is set to '365 ...	5	19012
Jun 26, 2026 @ 12:46:47.3...	Windows10-VM	CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Ensure 'Enforce password history' is set to '24 ...	5	19013
Jun 26, 2026 @ 12:46:45.1...	Windows10-VM	Discovery activity executed	3	92031
Jun 26, 2026 @ 12:46:45.1...	Windows10-VM	Discovery activity executed	3	92031
Jun 26, 2026 @ 12:46:41.3...	Windows10-VM	Discovery activity executed	3	92031

8.Ransomware Activity Simulation and Detection

To test the effectiveness of the SOC monitoring environment, a ransomware-like activity simulation was performed in the Windows Virtual Machine.

The objective of this simulation was to understand how a SOC analyst detects suspicious file activities, monitors endpoint behavior, and investigates security alerts.

Simulation Scenario

A suspicious file activity was generated on Windows VM to simulate ransomware behavior.

The monitoring system was configured to detect:

- Creation of suspicious files
- Unusual file modification activity
- Process execution behavior
- Changes in monitored directories
- Abnormal system activity

Detection Process

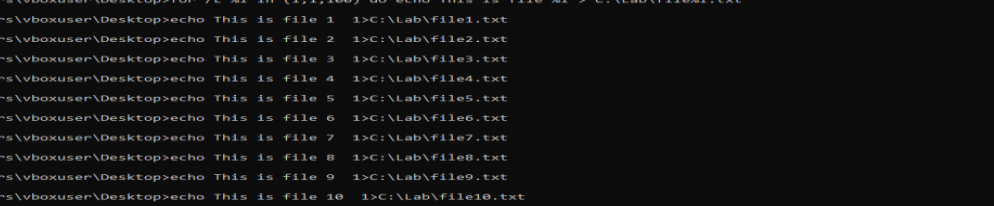
1. Suspicious file activity occurred on the Windows endpoint.
2. Sysmon captured the related system events.
3. Wazuh Agent collected the generated logs.
4. Wazuh Manager analyzed the events using configured detection rules.
5. An alert was generated on the Wazuh Dashboard.
6. The SOC analyst investigated the alert details.

Investigation Details

The following information was analyzed:

- Event timestamp
- Process name
- File path
- User account involved
- Source of activity
- Related security events

Evidence Screenshots



The screenshot shows a Windows Command Prompt window with the following text:

```

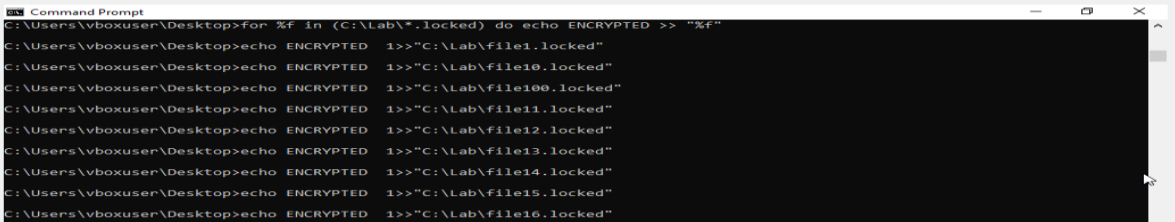
C:\Users\vboxuser\Desktop>for /L %i in (1,1,100) do echo This is file %i > C:\Lab\file%i.txt
C:\Users\vboxuser\Desktop>echo This is file 1 1>C:\Lab\file1.txt
C:\Users\vboxuser\Desktop>echo This is file 2 1>C:\Lab\file2.txt
C:\Users\vboxuser\Desktop>echo This is file 3 1>C:\Lab\file3.txt
C:\Users\vboxuser\Desktop>echo This is file 4 1>C:\Lab\file4.txt
C:\Users\vboxuser\Desktop>echo This is file 5 1>C:\Lab\file5.txt
C:\Users\vboxuser\Desktop>echo This is file 6 1>C:\Lab\file6.txt
C:\Users\vboxuser\Desktop>echo This is file 7 1>C:\Lab\file7.txt
C:\Users\vboxuser\Desktop>echo This is file 8 1>C:\Lab\file8.txt
C:\Users\vboxuser\Desktop>echo This is file 9 1>C:\Lab\file9.txt
C:\Users\vboxuser\Desktop>echo This is file 10 1>C:\Lab\file10.txt
C:\Users\vboxuser\Desktop>echo This is file 11 1>C:\Lab\file11.txt
C:\Users\vboxuser\Desktop>echo This is file 12 1>C:\Lab\file12.txt
C:\Users\vboxuser\Desktop>echo This is file 13 1>C:\Lab\file13.txt
C:\Users\vboxuser\Desktop>echo This is file 14 1>C:\Lab\file14.txt
C:\Users\vboxuser\Desktop>echo This is file 15 1>C:\Lab\file15.txt
C:\Users\vboxuser\Desktop>echo This is file 16 1>C:\Lab\file16.txt
C:\Users\vboxuser\Desktop>echo This is file 17 1>C:\Lab\file17.txt
C:\Users\vboxuser\Desktop>echo This is file 18 1>C:\Lab\file18.txt
C:\Users\vboxuser\Desktop>echo This is file 19 1>C:\Lab\file19.txt
C:\Users\vboxuser\Desktop>echo This is file 20 1>C:\Lab\file20.txt

```

```

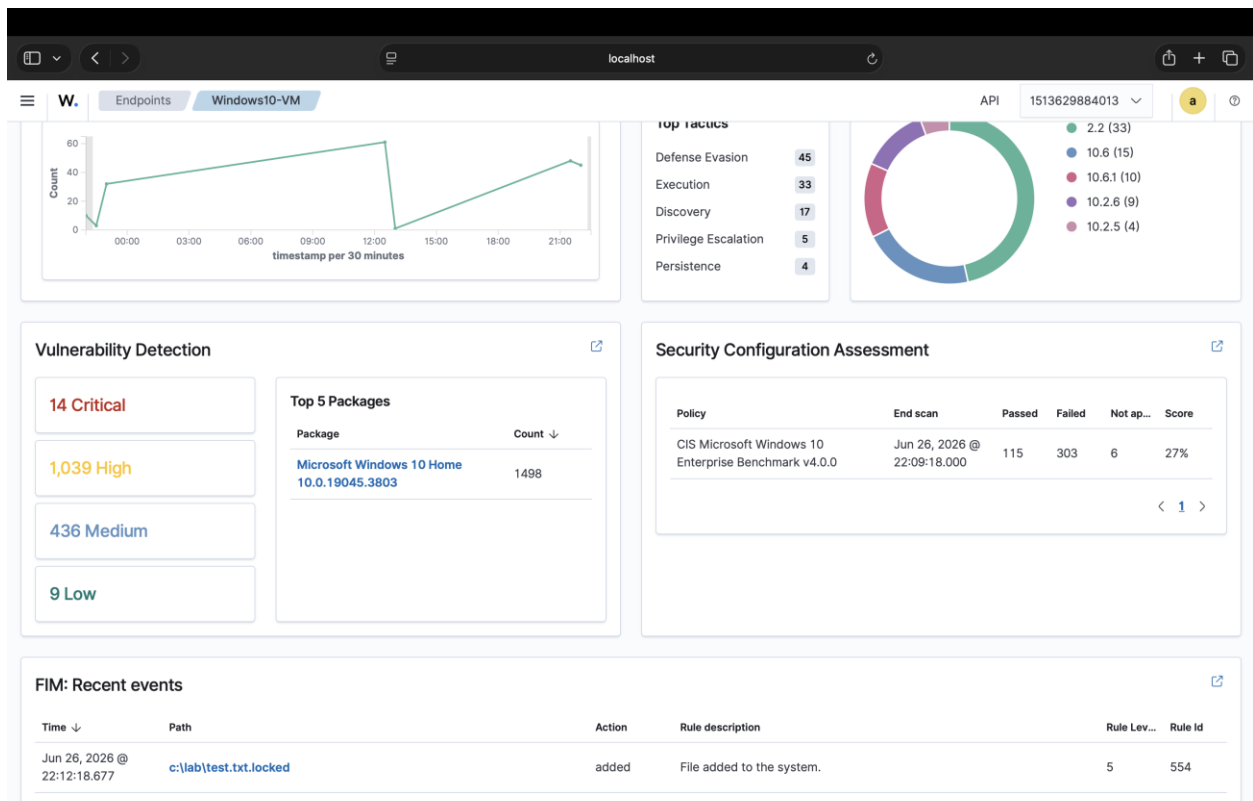
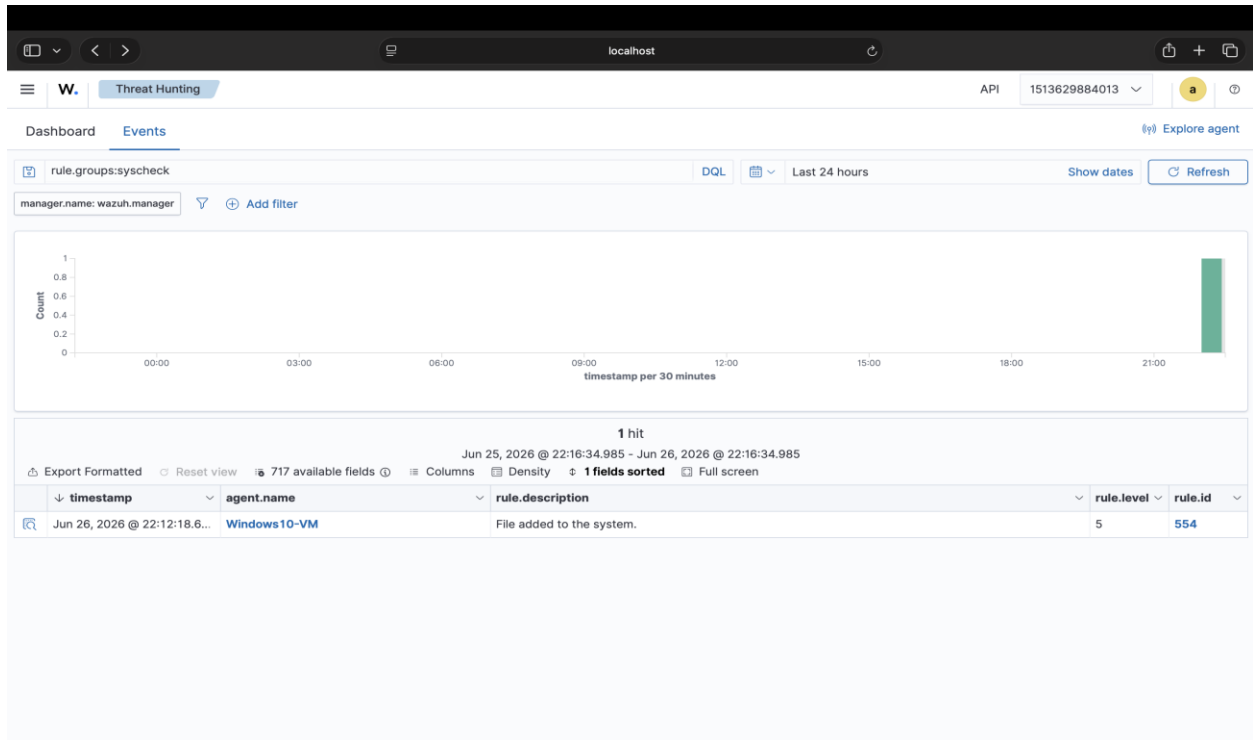
C:\Users\vboxuser\Desktop>for %f in (C:\Lab\*.txt) do ren "%f" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file1.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file10.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file100.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file11.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file12.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file13.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file14.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file15.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file16.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file17.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file18.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file19.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file2.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file20.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file21.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file22.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file23.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file24.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file25.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file26.txt" *.locked
C:\Users\vboxuser\Desktop>ren "C:\Lab\file27.txt" *.locked

```



```
C:\Users\vbouser\Desktop>for %f in (C:\Lab\*.locked) do echo ENCRYPTED >> "%f"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file1.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file10.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file100.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file11.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file12.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file13.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file14.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file15.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file16.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file17.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file18.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file19.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file2.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file20.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file21.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file22.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file23.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file24.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file25.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file26.locked"
C:\Users\vbouser\Desktop>echo ENCRYPTED 1>>"C:\Lab\file27.locked"
```


DETECTION :



9. Incident Analysis and Response Actions

After detecting the suspicious activity, the generated alert was analyzed to understand the impact and required response actions.

Incident Findings

The investigation identified:

- Suspicious file activity on the endpoint system
- Related process execution events
- Generated security alerts from Wazuh SIEM
- System activity recorded through Sysmon logs

Response Actions

The following incident response steps were performed:

1. Alert verification and analysis
2. Identification of affected system activity
3. Investigation of related logs
4. Monitoring of suspicious behavior
5. Documentation of findings

Incident Handling Process

The SOC workflow followed:

Detection → Analysis → Investigation → Response → Documentation

10. Project Results

The SOC Lab project successfully demonstrated a practical security monitoring environment.

The project achieved:

- Successful deployment of Wazuh SIEM using Docker
- Endpoint monitoring using Windows VM and Sysmon
- Log collection and security event analysis
- Creation of custom detection rules
- Detection and investigation of suspicious activities
- Understanding of SOC analyst workflow and incident response process

The project improved practical knowledge in Security Operations Center activities, SIEM monitoring, and Digital Forensics & Incident Response concepts.

11.Conclusion

This SOC Lab project provided hands-on experience in building and operating a basic Security Operations Center environment.

By integrating Docker, Wazuh SIEM, Windows VM, Sysmon, and custom detection rules, the project demonstrated how security events are collected, monitored, detected, and investigated.

The project helped in understanding real-world SOC operations including:

- Threat detection
- Log analysis
- Alert investigation
- Incident response
- Security monitoring

This practical implementation builds a foundation for further learning in cybersecurity, SOC operations, and Digital Forensics.

12. References

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<https://docs.docker.com/>
3. Microsoft Sysinternals – Sysmon Documentation
Microsoft Sysmon documentation was referred for understanding endpoint monitoring, event logging, and system activity tracking.
<https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon>
4. MITRE ATT&CK Framework
MITRE ATT&CK was referred for understanding adversary techniques, attack behaviors, and threat detection methodologies.
<https://attack.mitre.org/>
5. Windows Event Logging Documentation
Microsoft Windows event logging resources were referred for understanding security events and log analysis.
<https://learn.microsoft.com/en-us/windows/security/>
6. Wireshark Documentation
Wireshark documentation was referred for network traffic capture and packet analysis.
<https://www.wireshark.org/docs/>
7. Cybersecurity and Digital Forensics Concepts
Reference material was used for understanding SOC operations, incident response lifecycle, threat detection, and security monitoring practices.